

STRATEGY LAYER

1. A2 Dashboard & Exam Strategy

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A2 Format Breakdown

Section	Marks	Penalty	Questions
True / False	16	-½ wrong	16 Qs
MCQ (Multiple Choice)	44	-¼ wrong	22 Qs
Longer Questions	40	No penalty	~4–6 parts
Total	100	—	3 hours

Negative-Marking Decision Rule

For **TF**: Guess only if you can eliminate at least one option. Expected value of random guess = 0 ($1 \times \frac{1}{2} - \frac{1}{2} \times \frac{1}{2} = 0$). Guess if confidence > 55%.

For **MCQ**: With 4 options, random guess EV = $2 \times \frac{1}{4} - \frac{1}{4} \times \frac{3}{4} = +\frac{3}{16} > 0$. It pays to guess even randomly — but NEVER guess on TF if truly unsure.

80% Target = 80 marks

You need roughly 13/16 TF correct (assuming no wrong), 35/44 MCQ correct (with some wrong), and 32/40 on longer questions. Longer questions are your safest points — no penalty, and a good template gets you 60–70% even on unfamiliar topics.

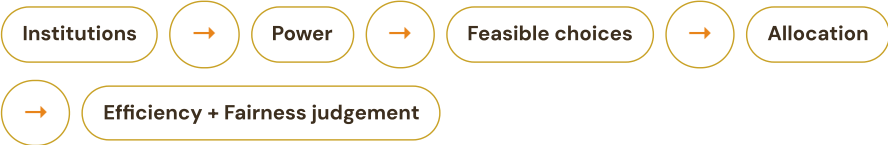
Source used in this pack	Type	Role
Economicstextbook Unit 1–13.docx	Primary text	All Angela/Bruno cases, definitions, data tables, MRS/MRT logic, inequality tools

Source used in this pack	Type	Role
2026 Econ 114 Unit 5CL – English.pdf	Lecture slides	Examinable wording, diagrams, case framing
2026 ECONOMICS 114 tut 5 Problem set.docx	Tutorial questions	AP/MP calculations, Peter Parker scenario, Jim/Dwight application
Slides_A2_correct_2.pptx	A2 revision deck	A2 format confirmation, scope (Units 4,5,6,7,8,10)

CONCEPTUAL FOUNDATION

2. Institutions, Power & Pirate Economics

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Institution (Textbook definition)

A set of **laws and informal rules** that regulate social interactions among people, and between people and the biosphere. Sometimes also called "the rules of the game". They determine who can do what, when they can do it, and what the pay-offs are.

Power (Two Forms)

Structural power: The value of your next best alternative (reservation option). Better outside option = more structural power = can walk away from bad deals.

Bargaining power: Ability to set the terms of exchange, make take-it-or-leave-it offers, or impose costs on the other party.

The Royal Rover: Pirate Economics

Captain Bartholomew Roberts' pirate ship had a *written constitution*. This shows that even illegal enterprises need institutions to coordinate behaviour and divide gains. The Royal Rover's articles were the rules of the game for that ship's economy.

Pirate Rule (Royal Rover)	Economic meaning	Exam lesson
Captain elected by majority vote	Limits unilateral power of one person	Institutions redistribute power and prevent exploitation

Pirate Rule (Royal Rover)	Economic meaning	Exam lesson
Quartermaster elected; can countermand captain outside battle	Checks and balances on authority	Power can be split by rules, not just held by one party
Lookout who spots a ship first receives "best pair of pistols"	Incentive scheme for effort	Rules shape behaviour through incentives
Compensation for combat injuries (specific body parts, specific amounts)	Risk-sharing and insurance	Institutions can promote fairness and cooperation
Multiracial crew with equal dividend rights	Equal property rights for members	Rules can reduce discrimination and inequality
Why can the same technology produce very different outcomes? ▼		
How do institutions affect the ultimatum game? ▼		

MEMORISATION LAYER

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3. Core Definitions Bank

Concept	Exam-ready definition (CORE 2.0)	Critical trap
Institution	A set of laws and informal rules that regulate social interactions among people, and between people and the biosphere — "the rules of the game"	NOT just "an organization". A university is an institution in common speech; in economics, the focus is on rules, not entities
Power	The ability to do and get things one wants in opposition to the intentions of others	Power can exist even in formally voluntary exchanges — voluntary ≠ equal power
Structural power	Power that derives from having a better reservation (fallback) option — the ability to walk away from a bad deal	Not the same as bargaining skill; it depends entirely on the next best alternative
Bargaining power	The ability to set terms, make take-it-or-leave-it offers, or impose costs on the other party	Bruno has bargaining power in Case 2 even though it is "voluntary"
Allocation	An outcome specifying who does what and who gets what — a complete description of the interaction result	Do not confuse with total production alone; the allocation specifies how it is shared
Economic rent	The benefit someone gets from a chosen action over and above the benefit of their next best alternative (reservation option)	Rent ≠ property rent; it means surplus above the fallback option
Reservation option	The next best alternative if an interaction does not happen; also called fallback option. Determines minimum acceptable pay-off	In Case 1 (coercion) Angela's reservation is escape/revolt. In Case 2 it is alternative employment — much better
Joint surplus	The sum of the economic rents of all parties in an interaction — the total gain above all parties' reservation options	Joint surplus can be large yet unequally shared; total ≠ fair distribution

Concept	Exam-ready definition (CORE 2.0)	Critical trap
Pareto improvement	A change that makes at least one person better off without making anyone else worse off	If even one person is made worse off, it is NOT a Pareto improvement regardless of how large the gains to others are
Pareto efficiency	An allocation is Pareto efficient if there is no feasible alternative allocation in which at least one person would be better off and nobody worse off	Pareto efficient does NOT mean fair, equal, or morally acceptable. Case 1 (forced labour) is Pareto efficient!
Substantive fairness	A judgement about an outcome based on the characteristics of the allocation itself — for example, whether incomes or wellbeing are equal	Focuses on what people get, not how they got it
Procedural fairness	A judgement about how an outcome came about — for example, whether it involved free choice, equal opportunity, or legitimacy	Focuses on the process and the rules, not the result
Lorenz curve	A graph showing the cumulative share of income or wealth received by the cumulative share of the population, ordered from poorest to richest	The curve shows distribution, NOT production. The further it bows below the 45° line, the more unequal
Gini coefficient	A numerical measure of inequality between 0 (perfect equality) and 1 (maximum inequality). Equals area between the Lorenz curve and the 45° line, divided by the total area under the 45° line	Higher Gini = more inequality. Gini = 0 means everyone has equal share; Gini ≈ 1 means one person has everything
Market income	Income before taxes and government transfers	Usually more unequal than disposable income; the comparison shows redistribution's effect
Disposable income	Income after taxes and transfers — what households actually have available to spend	Tax and transfer policy can lower the Gini of disposable income relative to market income

The Four-Quadrant Trap

In every Unit 5 answer, always keep four questions separate: (1) What is produced? (2) Who gets it? (3) Who had power and why? (4) Is the result Pareto efficient? Is it fair? Many wrong answers blend these together.

MODEL CORE

4. Angela/Bruno Model Foundations

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Angela — What she wants

Angela values **grain** (income/food) and **free time**. Her indifference curves are downward sloping: she is willing to give up some grain for more free time. Higher indifference curves = higher utility.

Key assumption (quasi-linear preferences): Angela's MRS depends only on her free time, NOT on how much grain she has. This means her indifference curves are vertical shifts of each other.

Bruno — What he wants

Bruno cares *only* about how much grain he receives. He does not farm; he owns the land. He does not care how much free time Angela has — only how much grain he gets. His indifference curves are **horizontal lines**.

His reservation option = 0 bushels (if Angela refuses/escapes). So any grain he receives is a positive economic rent.

Angela's Production Function — Textbook Data

Hours worked (h)	0	1	2	3	4	5	6	8	12	16	24
Grain output (bushels)	0	17	24	29	34	37	41	46	54	60	64
Free time (=24-h)	24	23	22	21	20	19	18	16	12	8	0

Reading the table:

At h=8 hours, Angela produces 46 bushels. Free time = 24-8 = 16 hours. This 8-hour / 46-bushel / 16-hour-free-time point appears in EVERY case as the optimum (because MRS=MRT always occurs at 16 hours free time in this model).

MRS — Marginal Rate of Substitution

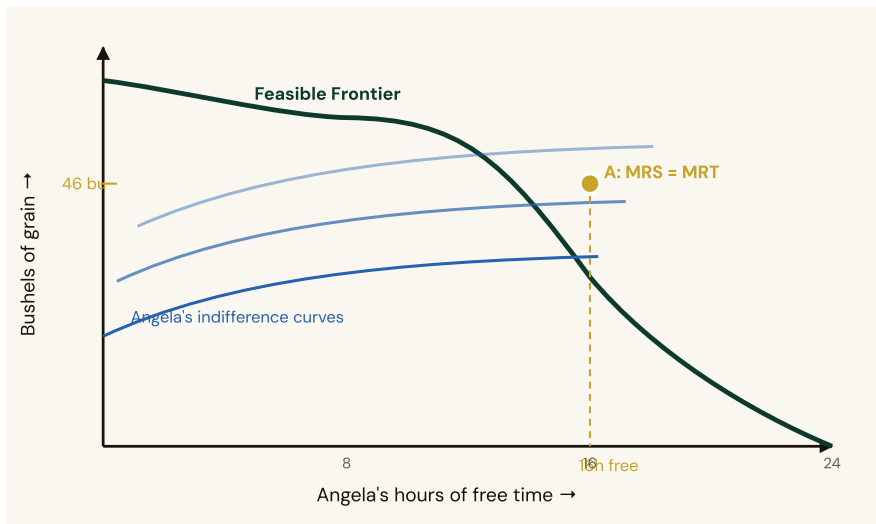
The slope of Angela's indifference curve (at any point). It represents: how many bushels of grain Angela is **willing to sacrifice** for one extra hour of free time.

MRS = how much grain she values free time (willingness to trade)

MRT — Marginal Rate of Transformation

The slope of the feasible frontier (at any point). It represents: how much grain Angela must sacrifice to gain one more hour of free time (= the marginal product of an hour of work).

MRT = opportunity cost of free time in terms of grain foregone



The feasible frontier (from the production function) and Angela's indifference curves. The optimum is at A (16 hours free, 46 bushels) where the curves are tangent: $MRS = MRT$.

// Key relationship — always true at any optimum in this model:

MRS = MRT ← at the optimal point

// MRS = slope of IC (willingness to trade free time for grain)

// MRT = slope of feasible frontier (opportunity cost of free time in grain)

// Why: if $MRT > MRS$, taking 1 less hour free gives more grain than Angela requires to stay equally well off

// → Angela (or Bruno) can do better by reducing free time

// If $MRS > MRT$, Angela can increase utility by taking more free time

BASELINE CASE

5. Independent Farmer — Baseline

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In the baseline case, Angela **owns the land** herself. There is no Bruno. She chooses her hours to maximise her utility, keeping all the grain she produces. This is a one-person constrained choice problem.

Feature	Detail
Institution	Private property — Angela owns land and its produce. Government enforces her right to exclude others.
Who chooses hours?	Angela alone — she maximises utility on the feasible frontier
Decision rule	Choose point where IC is tangent to feasible frontier: MRS = MRT
Outcome	16 hours free time; works 8 hours; produces and consumes 46 bushels
Bruno's share	n/a — Bruno is not a character in this case
Angela's economic rent	Positive — she is better off than in any other case
Pareto efficiency	Yes — she is at her optimum; no change can make her better off without consuming more than is feasible
Fairness	No distributional conflict — only one person

Exam Wording (use this):

"As an independent farmer, Angela chooses the combination of grain and free time that maximises her utility subject to the feasible frontier. The optimum occurs where her MRS (willingness to trade free time for grain) equals the MRT (opportunity cost of free time). At this point she has 16 hours of free time, works 8 hours, and consumes 46 bushels of grain."

Why does Angela not choose to have 24 hours of free time and produce 0 grain? ▼

CASE 1

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6. Forced Labour — Coercion

Bruno owns the land AND can coerce Angela physically. The government enforces his right to use force. Angela's only option is to comply or attempt to escape/revolt (risky and costly). Her **reservation indifference curve IC1** represents her minimum acceptable utility under threat of force.

Reservation Indifference Curve IC1

Each point on IC1 gives Angela exactly the utility she would get from disobeying Bruno and risking escape or revolt. Bruno must keep Angela on or above IC1 to prevent resistance. Below IC1 = Angela revolts.

Bruno's Feasible Set

The lens-shaped region between the **feasible frontier** (above) and IC1 (below). All allocations here are feasible AND will be complied with. Bruno maximises his grain within this region.

How Bruno Maximises — The Logic

Bruno keeps Angela exactly on IC1 (giving her no economic rent). The grain he captures is the vertical distance between the feasible frontier and IC1. He maximises this vertical distance by choosing where:

Optimal condition for Bruno under coercion:

MRT (slope of feasible frontier) = MRS (slope of IC1 at that point)

// To the left (Angela works more): $MRS > MRT \rightarrow$ Bruno should give her MORE free time

// To the right (Angela works less): $MRS < MRT \rightarrow$ Bruno should give her LESS free time

// Optimum is at 16 hours free time (same as Angela's own choice!)

Bruno's rent = Total grain - Angela's survival grain
= 46 - 15 = 31 bushels

Item	Baseline (Angela owns)	Case 1 (Coercion)
Who decides hours?	Angela (freely)	Bruno (by force)
Angela's free time	16 hours	16 hours
Angela's grain	46 bushels	15 bushels
Bruno's grain	0 (not present)	31 bushels
Angela's economic rent	Positive	ZERO (on IC1)
Bruno's economic rent	n/a	31 bushels
Pareto efficient?	Yes	YES — surprisingly!
Fair?	n/a	Very unfair — coercion, no consent

Why is Case 1 Pareto efficient?

Consider any change from allocation D. If Angela gets more grain, Bruno is worse off. If hours change, Bruno's maximum grain decreases. If Angela is pushed below IC1, she revolts and both get nothing. Therefore no change can make one person better off without making another worse off $\rightarrow$ Pareto efficient. *This is the most counterintuitive result in Unit 5 — learn it word-for-word.*

Exam Trap:

"If Angela survives, the outcome must be fair." FALSE. Survival only means Bruno gave her IC1 — it says nothing about fairness, equality, or consent. An allocation can be efficient AND deeply unfair.

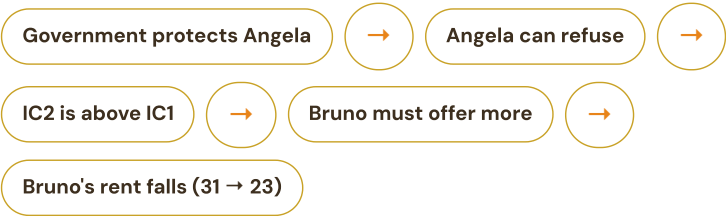
Why does Angela work the same hours in Case 1 as in the baseline?

CASE 2

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7. Take-It-or-Leave-It Contract

In Case 2, Bruno owns the land but **cannot use physical force**. The government protects Angela from coercion and enforces contracts. Angela can refuse and seek work elsewhere — her reservation option is much better than in Case 1. But Bruno still has **bargaining power**: he makes the take-it-or-leave-it offer.



Employment Contract

Bruno specifies hours (8) and wage (23 bushels). Angela accepts if wage ≥ IC2. Bruno makes a take-it-or-leave-it offer at minimum acceptable wage, capturing all joint surplus.

Condition: Bruno offers a contract such that Angela is exactly on IC2 (reservation utility). MRS on IC2 = MRT at optimum → same 16 hours free time, 8 hours work.

Tenancy Contract

Bruno specifies a **land rent of 23 bushels**. Angela is free to choose her own hours. She solves her own optimum with consumption = output - 23. She ends up choosing 8 hours work (same as before) and consuming 23 bushels.

The outcome is identical — Bruno gets the same 23 bushels whether he sets hours directly (employment) or sets a land rent (tenancy).

Item	Case 1 (Coercion)	Case 2 (Take-it-or-leave-it)
Angela's reservation option	Escape/revolt (very risky)	Alternative employment (much better)
Angela's IC	IC1 (lower)	IC2 (higher)

Item	Case 1 (Coercion)	Case 2 (Take-it-or-leave-it)
Angela's free time	16 hours	16 hours (same)
Angela's grain	15 bushels	23 bushels (better)
Bruno's grain	31 bushels	23 bushels (less)
Joint surplus	46 bushels	46 bushels (same total)
Angela's economic rent	0	STILL 0 (Bruno has take-it-or-leave-it power)
Bruno's economic rent	31	23 (= entire joint surplus above IC2)
Pareto efficient?	Yes	Yes
Fair (substantive)?	Very unfair (31 vs 15)	More equal (23 vs 23) but Angela gets 0 rent

Key insight on joint surplus:

The joint surplus is maximised at MRS = MRT → 23 bushels (= 46 total - 23 for Angela's IC2). Since Bruno makes the take-it-or-leave-it offer, he captures the *entire* joint surplus as rent. Angela gets exactly her reservation utility, zero rent. Voluntary exchange does not mean equal surplus sharing.

Trap: "Angela must be better off in Case 2 than Case 1."

TRUE for Angela (15 → 23 bushels). FALSE for Bruno (31 → 23). The Pareto criterion cannot rank Case 1 vs Case 2: Angela prefers Case 2, Bruno prefers Case 1. Neither Pareto dominates the other.

Why does the tenancy contract produce the same outcome as the employment contract? ▾

CASE 3

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8. Bargaining in a Democracy

In Case 3, Angela and other farmers can **vote**. They elect a government that passes legislation: a maximum working week and minimum wage equal to what Angela would earn in Case 2. Now Angela has **both structural power** (better reservation option) and some **bargaining power** (she and Bruno can negotiate, not just accept/reject).

Step 1: Legal Minimum

The law sets: maximum 4.5 hours work, minimum wage = 23 bushels (same as Case 2). This moves Angela to a point above IC2 — she gets the same 23 bushels for *fewer hours work* (19.5 hours free instead of 16). Angela is better off than in Case 2.

Step 2: Further Bargaining

At the legal minimum, **MRS ≠ MRT**. Both Angela and Bruno can gain by renegotiating: Angela works slightly more, the extra grain is shared. This is a Pareto improvement over the legal minimum.

Why is the legal minimum NOT automatically Pareto efficient?

Because it restricts work to 4.5 hours, not 8 hours. At 4.5 hours work (19.5 hours free), $MRT > MRS$: the grain that could be produced by one extra hour of work exceeds the grain Angela would need to compensate her for losing that free time. So there is a potential mutual gain from increasing hours and sharing the extra grain. The parties are not at the $MRS = MRT$ point — a Pareto improvement is still possible.

Item	Case 2 (Take-it-or-leave-it)	Case 3 Legal Minimum	Case 3 After Bargaining
Institution	Property rights only	Labour law + property rights	Labour law + negotiation
Angela's free time	16h	19.5h (more!)	Between 16h and 19.5h
Angela's grain	23	23	>23 (more rent)

Item	Case 2 (Take-it-or-leave-it)	Case 3 Legal Minimum	Case 3 After Bargaining
Bruno's grain	23	Less than 23	Depends on bargain
Angela's economic rent	0	Positive	Positive
Pareto efficient?	Yes	NO (MRS ≠ MRT)	YES (if MRS = MRT reached)

How can both Bruno and Angela be better off after bargaining vs the legal minimum?▼

EVALUATION LAYER

9. Pareto Efficiency & Fairness — The Core Tension

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The critical distinction Unit 5 tests every year:

An allocation can be *Pareto efficient and deeply unfair at the same time*. Efficiency asks whether any mutual gain is still possible. Fairness asks whether the distribution is just, equal, or legitimate.

Pareto Efficiency

An allocation is Pareto efficient if there is NO feasible alternative in which at least one person is better off and nobody is worse off.

- Does NOT require equality
- Does NOT require consent or fairness
- A brutal dictatorship can be Pareto efficient
- Case 1 (forced labour) IS Pareto efficient

Fairness (Two Types)

Substantive: Judged by what people get — income, wellbeing, freedom. Equal pay for equal work. Different criteria lead to different conclusions.

Procedural: Judged by how the outcome came about — free choice, equal opportunity, legitimacy, no force. Case 1 fails on procedural grounds even if both survive.

Rawls' Veil of Ignorance

The philosopher John Rawls argued for evaluating institutions from behind a "veil of ignorance" — not knowing what position you would occupy in society (rich/poor, Angela/Bruno). From this impartial position, you evaluate the rules of the game. Would you accept the rules of forced labour if you did not know whether you would be Angela or Bruno?

High-yield sentence (copy in exam answers):

"An allocation can be Pareto efficient and still be unfair, because Pareto efficiency only asks whether any mutual gains remain, not whether the distribution of power, opportunity, or surplus is just."

MASTER SUMMARY

10. Master Case Comparison Table

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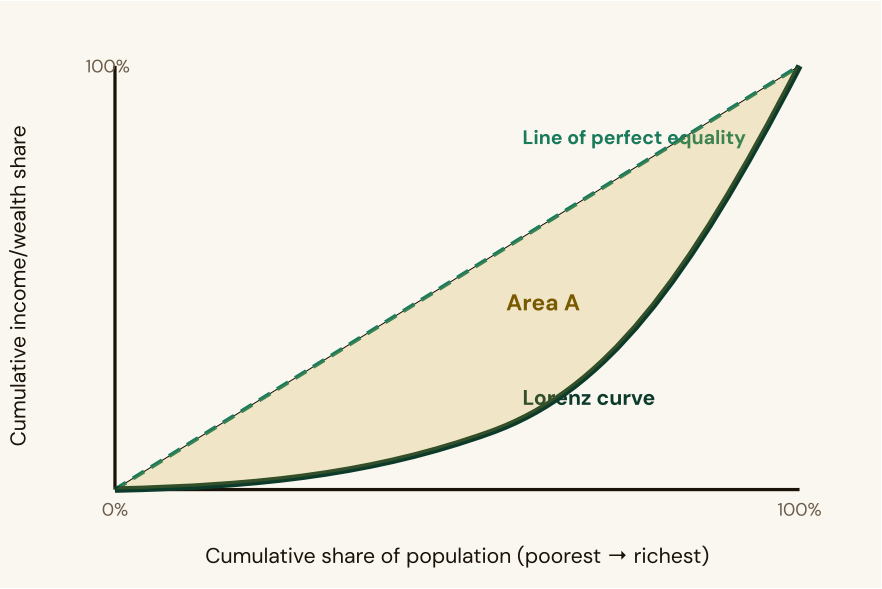
Dimension	Baseline	Case 1: Coercion	Case 2: Take-it-or-leave-it	Case 3: Democracy + Bargaining
Institution	Angela owns land	Bruno owns + can coerce	Bruno owns; no coercion; contracts enforced	Bruno owns; labour law; bargaining possible
Reservation option (Angela)	Own production (best)	Escape/revolt (worst)	Alternative employment	Legal minimum wage
Who decides hours?	Angela	Bruno	Bruno (employment) or Angela (tenancy)	Legal limit + bargaining
Angela's grain	46	15	23	>23
Bruno's grain	0 (absent)	31	23	<23
Angela's free time	16h	16h	16h	More than 16h
Angela's economic rent	Positive (maximum)	Zero	Zero	Positive
Bruno's economic rent	0	31 bushels	23 bushels	Positive but less than 23
Pareto efficient?	Yes	Yes (!)	Yes	Legal min: NO; After bargain: YES

Dimension	Baseline	Case 1: Coercion	Case 2: Take-it-or-leave-it	Case 3: Democracy + Bargaining
Substantively fair?	n/a	Very unfair (15 vs 31)	More equal (23 vs 23) but process unfair	More fair (Angela gets rent)
Procedurally fair?	Yes	No (coercion)	Debatable (voluntary but unequal power)	More legitimate (democratic rules)
Exam trap	Introducing Bruno here	Saying coercion is always inefficient	Saying voluntary = fair	Saying legal minimum is Pareto efficient

INEQUALITY TOOLS

11. Lorenz Curve & Gini Coefficient

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Gini = Area A + (Area A + Area B under the equality line). A curve bowing further from the equality line = higher Gini = more inequality.

Tool	How to construct/read	Limits / range	Exam trap
Lorenz curve	Order population from poorest to richest. Plot cumulative % of income/wealth on y-axis vs cumulative % of population on x-axis.	Always starts at (0,0) and ends at (100%,100%)	It is a distribution tool, NOT a production tool. Never use it to show output.
45° equality line	The line where every x% of the population earns exactly x% of income. Perfect equality.	A straight diagonal from origin to (100%,100%)	Do NOT call it the "efficient" line. It is the equality benchmark.

Tool	How to construct/read	Limits / range	Exam trap
Gini coefficient	= Area A + (Area A + B). Calculated from the gap between the equality line and the Lorenz curve.	0 = perfect equality; 1 = maximum inequality (one person has everything)	Higher Gini ≠ more poverty; it measures distribution not absolute living standards
Market income Gini	Calculated before taxes and transfers	Usually higher (more unequal)	The market Gini is not the policy-relevant measure alone
Disposable income Gini	Calculated after taxes and transfers	Usually lower	The gap between market and disposable Gini shows redistribution impact

Can two countries have the same Gini but very different living standards? ▼

POLICY APPLICATIONS

12. Redistribution & Policy Applications

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Redistribution via Income

Taxes and transfers: progressive income tax + transfers (welfare, grants) reduce market income inequality. Market income Gini > Disposable income Gini after redistribution.

Redistribution via Rights

Labour laws, voting rights, property rights: change bargaining power *before* income is even distributed. Case 3 shows how democratic rights shift the balance without explicit tax transfers.

Redistribution via Policy Design

Policies can affect both the size of the surplus AND who gets it. Sometimes removing a power imbalance that prevents efficient bargaining can simultaneously improve fairness and efficiency.

Application	Conflict	Institutional question	Unit 5 analysis
Wages	Workers want higher pay; employers want lower costs	Are unions legal? Are minimum wages set? Can workers credibly threaten to quit?	Institutions change structural and bargaining power → affects surplus division. Minimum wage raises IC/reservation option.

Application	Conflict	Institutional question	Unit 5 analysis
Pollution	Firms benefit from production; nearby communities bear health costs	Who has the legal right to clean air/water? Can affected communities negotiate compensation?	Property rights over environment = power to impose costs without consent. Legal right to sue = bargaining power for affected parties.
Jobs vs regulation	Regulation may protect workers but raise firm costs	What rights do workers have? Who bears adjustment costs?	Evaluate both Pareto efficiency (does regulation prevent mutually beneficial exchanges?) and fairness (who bears risk?).

Longer-question warning:

Avoid vague moral language like "this is unfair" or "this is bad for society." Convert fairness into economics: who has power, what is the reservation option, who receives surplus, was the process voluntary, are property rights legitimate?

VISUAL MASTERY

13. Graph Bank — 7 Diagrams to Master

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Diagram	Axes	Key feature	Exam sentence	Common mistake
Production function	x: hours worked; y: grain output	Concave (diminishing MP). Slope at any point = MP	"The slope of the production function at any point is the marginal product of that hour of work; it diminishes because land is fixed."	Confusing hours worked (x-axis) with hours free time (x-axis of feasible frontier)
Feasible frontier	x: free time; y: grain	Mirror image of production function. Downward sloping. Slope = MRT	"Moving right along the feasible frontier increases free time but reduces grain production."	Forgetting that free time = 24 – hours worked. Free time is not the same as leisure.
Angela's indifference curves	x: free time; y: grain	Downward sloping, quasi-linear (parallel vertical shifts). Slope = MRS	"Higher indifference curves represent higher utility. MRS flattens as free time increases (diminishing MU of free time)."	Making ICs steeper than the feasible frontier everywhere, missing the tangency point
Case 1 — coercion feasible set	x: free time; y: grain	Lens-shaped region between feasible frontier	"The lens-shaped area shows all	Drawing IC1 above the feasible frontier

Diagram	Axes	Key feature	Exam sentence	Common mistake
		and IC1. Bruno's feasible set.	allocations Bruno can enforce. He maximises grain at the point where IC1 is tangent to the feasible frontier."	(infeasible) or below the origin
Case 2 — take-it-or-leave-it	x: free time; y: grain	IC2 above IC1; allocation L on IC2 at MRS=MRT; Bruno gets vertical gap between feasible frontier and IC2	"Allocation L gives Angela her reservation utility (IC2). Bruno captures the entire joint surplus as the vertical gap between the feasible frontier and IC2."	Putting IC2 below IC1 (incorrect — better outside options → higher IC)
Pareto efficiency curve	x: free time; y: grain allocation	Vertical line at 16 hours free time (because of quasi-linear preferences). All efficient allocations are on this line.	"In this model, all Pareto-efficient allocations share the same work hours (8 hours) but differ in how grain is divided between Angela and Bruno."	Thinking there is only one Pareto-efficient allocation; in fact there is a range (all with the same hours)
Lorenz curve	x: cumulative % of population; y: cumulative % of income/wealth	Bows below 45° line. Area A between curve and equality line divided by total = Gini	"The further the Lorenz curve bows below the 45° line, the larger area A, and the higher	Using it to show production levels instead of income/wealth distribution

Diagram	Axes	Key feature	Exam sentence	Common mistake
			the Gini coefficient."	

QUANTITATIVE LAYER

14. Formula, Calculation & Interpretation Bank

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Formula/Concept	Definition	Worked Example (Angela's data)	Interpretation	Common error
Average Product (AP)	AP = Y / h Output per hour of work	h=8, Y=46: AP = 46/8 = 5.75 bushels/hour	On average each hour of work produces 5.75 bushels	AP ≠ MP. AP is average, MP is the marginal (extra) unit
Marginal Product (interval)	MP ≈ ΔY / Δh Extra output per extra hour	h: 6→8, Y: 41→46: MP = (46-41)/(8-6) = 5/2 = 2.5 bu/hr	Each additional hour between h=6 and h=8 produces about 2.5 extra bushels	This is an interval approximation. MP at a specific point requires calculus
Marginal Product (calculus)	MP = dY/dh Derivative of production function	If Y = 5h - 0.1h², then MP = 5 - 0.2h At h=10: MP = 5 - 2 = 3	The 10th hour of work adds 3 bushels to output	Only use derivative form if a functional form is given in the question
Bruno's rent (coercion)	Bruno's grain = Total grain - Angela's minimum grain i.e., residual above IC1	Total = 46, Angela needs 15: Bruno = 46 - 15 = 31 bushels	Bruno captures 31 out of 46 bushels — 67% of output	The survival minimum is NOT zero grain — Angela needs grain to stay alive and work
Joint surplus (Case 2)	JS = Total grain - Angela's IC2 grain requirement Vertical gap	Total = 46, IC2 requires 23: JS = 46 - 23 = 23 bushels	23 bushels is available above Angela's reservation utility; Bruno	Joint surplus depends on Angela's reservation option — it is

Formula/Concept	Definition	Worked Example (Angela's data)	Interpretation	Common error
	between feasible frontier and IC2		captures all of it in Case 2	larger in Case 1 (lower IC)
Gini coefficient	Gini = Area A / (Area A + B) From the Lorenz diagram	If Area A = 0.2 and total area under 45° line = 0.5: Gini = 0.2/0.5 = 0.40	A Gini of 0.40 means moderate inequality. South Africa has one of the world's highest at ~0.63	Gini = 0 ≠ poverty; it means perfect equality of whatever is being measured

Tutorial 5 Calculation Practice (from problem set)

Given $ATP = 16,000 / (h + 32) + 2h$:
Q4.1: AP at $h = 25$: $ATP = 16000 / 57 + 50 \approx 280.7 + 50 =$
330.7

Q4.2: Total output $Q = h \cdot ATP = 16000h / (h + 32) + 2h^2$. $MP = dQ/dh = 512000 / (h + 32)^2 + 4h$
Q4.3: MP at $h = 105$: $= 512000 / (137)^2 + 420 = 512000 / 18769 + 420 \approx 27.3 + 420 =$
~447

TF PRACTICE — NEGATIVE MARKING

15. True / False Bank (20 Questions)

0% complete

Click on a question to reveal the answer. Practice saying TRUE or FALSE before you click.

Q1. In Case 1, Angela receives no economic rent. The outcome is Pareto efficient. ▲

TRUE.
Angela is on IC1 with zero rent above her reservation option. The outcome IS Pareto efficient — any change makes at least one party worse off. Case 1 shows that efficiency and fairness are completely different criteria.

Q2. In Case 2 (take-it-or-leave-it), the exchange is voluntary, therefore both parties receive an equal share of the joint surplus. ▲

FALSE.
Voluntariness does not imply equal surplus sharing. Since Bruno makes the take-it-or-leave-it offer, he captures the entire joint surplus (23 bushels in the example). Angela receives zero economic rent — she is exactly at her reservation utility.

Q3. Angela's MRS at 16 hours of free time is the same on every indifference curve. ▲

TRUE.
This is the quasi-linear preference assumption. The slope of Angela's ICs at any given free-time level is the same regardless of how much grain she has. This is why Angela always works 8 hours at the optimum, regardless of which case we are in.

Q4. The Lorenz curve shows how much output is produced at different levels of employment. ▲

FALSE.

The Lorenz curve shows the distribution of income or wealth, NOT production levels. It plots cumulative income share against cumulative population share. Do not confuse it with a production function.

Q5. An allocation that is Pareto efficient must also be substantively fair. ▲

FALSE.

This is THE most important Unit 5 trap. Case 1 (forced labour, Angela gets 15, Bruno gets 31) is Pareto efficient but substantively extremely unfair. Pareto efficiency only says no mutual gains remain — it says nothing about equality or justice.

Q6. Under a tenancy contract in Case 2, Angela freely chooses how many hours to work and ends up working 8 hours. ▼

Q7. In Case 1, Bruno would maximise his grain by forcing Angela to work 24 hours. ▼

Q8. The feasible frontier is the mirror image of the production function, with free time on the horizontal axis instead of hours worked. ▼

Q9. A Gini coefficient of 0 means everyone is equally poor. ▼

Q10. In Case 3, the initial legal minimum (maximum hours law) is Pareto efficient. ▼

Q11. Structural power depends on the quality of a person's next best alternative to the current interaction. ▼

Q12. The joint surplus is maximised at the same point regardless of which case we are in (Case 1, 2, or 3). ▼

Q13. Procedural fairness asks whether the outcome (allocation) is equal or just. ▼

Q14. Under a take-it-or-leave-it contract, Angela's working hours are necessarily the same as under forced labour. ▼

Q15. Market income Gini is always lower than disposable income Gini. ▼

Q16. Bargaining power and structural power are the same thing. ▼

Q17. Rawls' veil of ignorance is used to evaluate the Pareto efficiency of an allocation. ▼

Q18. In the Angela/Bruno model, Bruno's indifference curves are horizontal lines. ▼

Q19. If Angela's biological survival constraint shifts upward, Bruno will be able to extract more grain from Angela. ▼

Q20. An employment contract and a tenancy contract always produce the same outcome for Angela and Bruno. ▼

MCQ PRACTICE

16. Multiple Choice Question Bank (20 Questions)

0% complete

MCQ 1

Angela produces grain on Bruno's land. In the Angela/Bruno model, the feasible frontier shows:

- A. The maximum grain producible for each combination of labour and capital
- B. Angela's preferences between grain and free time
- C. The maximum grain Angela can produce for each amount of free time she takes, given her technology
- D. The minimum grain Angela needs to survive at each work level

MCQ 2

Under Case 1 (forced labour), Bruno maximises his grain at the allocation where:

- A. Angela's total output is at its maximum ($h=24$ hours)
- B. The slope of the feasible frontier equals the slope of Angela's reservation indifference curve IC1
- C. Angela receives zero grain
- D. Bruno's grain is equal to Angela's grain

MCQ 3

An allocation is said to Pareto dominate another if:

- A. It produces more total output
- B. It is fairer according to Rawls' veil of ignorance

C. At least one person is better off and nobody is worse off compared to the original

D. Both Angela and Bruno receive more grain than before

MCQ 4

Compared to Case 1, in Case 2 (take-it-or-leave-it), which of the following is true?

- A. Angela works more hours, so total grain is higher
- B. Angela is better off and Bruno is also better off
- C. Angela is better off but Bruno is worse off; neither allocation Pareto dominates the other
- D. Bruno captures more of the joint surplus because the market is competitive

MCQ 5

Which of the following correctly describes Angela's indifference curves in the Angela/Bruno model?

- A. Horizontal lines, because Angela only cares about grain
- B. Downward-sloping curves that become flatter as free time increases (quasi-linear)
- C. Upward-sloping curves because more work produces more grain
- D. Vertical lines, because Angela's utility depends only on her free time

MCQ 6

The Gini coefficient of disposable income is usually lower than the Gini coefficient of market income because:

- A. People with low market income tend to work harder
- B. Taxes and government transfers reduce income inequality between households
- C. The Lorenz curve for disposable income bows outward from the equality line

D. Market income includes capital gains which are always positive

MCQ 7

Angela works 8 hours and produces 46 bushels. What is her average product?

- A. 8 bushels per hour
- B. 5.75 bushels per hour
- C. 46 bushels per hour
- D. 368 bushels

MCQ 8

Using the production function table (h=6: 41 bushels; h=8: 46 bushels), what is the approximate marginal product of labour between h=6 and h=8?

- A. 5 bushels per hour
- B. 2.5 bushels per hour
- C. 41 bushels per hour
- D. 23 bushels per hour

MCQ 9

In Case 3 (democracy and bargaining), the initial legal minimum (restricting Angela's work to fewer hours) is not Pareto efficient because:

- A. Angela is now worse off than in Case 2
- B. MRT exceeds MRS at the restricted work level, so there are still gains available from further bargaining
- C. Bruno refuses to accept the legal constraint and uses force instead
- D. The Gini coefficient rises after the law is passed

MCQ 10

Which evaluation criterion focuses on the characteristics of the outcome itself (e.g., income equality) rather than how it came about?

- A. Procedural fairness
- B. Pareto efficiency
- C. Substantive fairness
- D. Structural power

MCQ 11

Which of the following would INCREASE Bruno's structural power?

- A. Angela gains the right to vote
- B. A new law sets a minimum wage for farm workers
- C. Angela's alternative employment options worsen (fewer jobs available)
- D. Bruno's land becomes less productive

MCQ 12

The Lorenz curve for Country A is closer to the 45° equality line than the Lorenz curve for Country B. Which is correct?

- A. Country A is wealthier than Country B
- B. Country A has a higher Gini coefficient than Country B
- C. Country A has lower income inequality than Country B
- D. Country A has a larger population than Country B

MCQ 13

Why is Case 1 (forced labour) Pareto efficient, despite being deeply unfair?

- A. Because Angela freely chose the allocation

B. Because any change from allocation D would make at least one party worse off

C. Because Angela receives more grain than Bruno

D. Because the total output is at its maximum possible level

MCQ 14

In a take-it-or-leave-it contract (Case 2), Angela receives zero economic rent because:

A. She is not working hard enough to earn a rent

B. Bruno uses force to prevent Angela from earning more

C. Bruno sets the terms so that Angela receives exactly her reservation utility — her rent above her fallback is zero

D. The total output is insufficient to pay Angela any rent

MCQ 15

The slope of the feasible frontier at any point represents:

A. Angela's MRS — her willingness to trade grain for free time

B. Bruno's indifference curve slope

C. The MRT — the marginal product of labour (opportunity cost of free time in grain)

D. The average product at that level of free time

MCQ 16

Which of the following best describes the role of institutions in the Angela/Bruno model?

A. They determine Angela's preferences and the shape of her indifference curves

B. They determine the production technology and the shape of the feasible frontier

C. They determine who has power, what options are feasible, and how the surplus is divided

D. They fix the price of grain in the market

MCQ 17

What is economic rent for a person engaged in an economic interaction?

A. The payment made to use someone else's property

B. The benefit received from a chosen action above the benefit of the next best alternative

C. The total amount of grain a person receives in the Angela/Bruno model

D. A government subsidy paid to farm workers

MCQ 18

If the government introduces a minimum wage equal to Angela's Case 2 wage (23 bushels) in Case 3, which statement is correct?

A. Angela is now worse off than in Case 2

B. Angela receives the same grain as in Case 2 but with less work, making her better off

C. Bruno is better off because the minimum wage increases total output

D. The outcome is now Pareto efficient because $MRS = MRT$ at the legal minimum

MCQ 19

Which of the following is a substantive judgement of fairness?

A. "The contract was voluntary, so the outcome is fair"

B. "The captain was democratically elected, making his orders legitimate"

C. "Angela receives only 15 bushels while doing all the work — this distribution is unequal and unfair"

D. "The rules of the game apply equally to everyone"

MCQ 20

If Angela's production function output at $h=12$ is 54 bushels and at $h=8$ is 46 bushels, what is the average marginal product over this range?

A. 54 bushels per hour

B. 4 bushels per hour

C. 2 bushels per hour

D. 8 bushels per hour

LONGER QUESTIONS — 40 MARKS

0% complete

17. Longer Question Templates

Template 1: "Compare two institutional cases" (6–10 marks)

- Define** the key concept: "An institution is a set of laws and informal rules — the rules of the game. In Case [X], the institution specifies that..."
- Identify the model:** "In the Angela/Bruno model, Angela produces grain and Bruno owns the land. The institution determines who has power and what each party can get."
- Explain Case X mechanism:** "Under Case X, Bruno can/cannot [use force / make a take-it-or-leave-it offer / negotiate]. Angela's reservation option is [IC1/IC2/legal minimum]."
- State the outcome:** "At the optimum ($MRS = MRT$), Angela has 16 hours free and produces 46 bushels. Of this, Angela receives [X] and Bruno receives [Y]."
- Compare to Case Y:** "Compared to Case Y, Angela is [better/worse] off because [better reservation option / same bargaining power]. Bruno is [better/worse] off because [higher/lower rent]."
- Evaluate:** "The allocation in Case X is Pareto [efficient/inefficient] because [no/yes] change could improve both parties. It is [fair/unfair] on substantive grounds because [income distribution] and [fair/unfair] on procedural grounds because [voluntary/coerced]."

Template 2: "Is an allocation efficient and/or fair?" (8–12 marks)

- Define Pareto efficiency:** "An allocation is Pareto efficient if there is no feasible alternative in which at least one person is better off and nobody is worse off."
- Check efficiency using MRS/MRT:** "At the proposed allocation, if $MRS [= / \neq] MRT$, then [no/a] Pareto improvement is possible because [the slopes are balanced / changing hours and sharing grain can make both better off]."
- Apply Pareto criterion to specific change:** "If we move from this allocation to [alternative], [Angela/Bruno] would be [better/worse] off by [amount]. Because [person] is worse off, this is not a Pareto improvement."

- 4 **Evaluate substantive fairness:** "On substantive grounds, the allocation gives Angela [X] and Bruno [Y]. This is [equal/unequal] and may be judged [fair/unfair] if we measure fairness by [income equality / wellbeing]."
- 5 **Evaluate procedural fairness:** "On procedural grounds, the outcome arose from [voluntary exchange / coercion / democratic legislation]. The process [was / was not] fair because [equal opportunity / free choice / legitimate rules]."
- 6 **Conclude with the key tension:** "This illustrates that Pareto efficiency and fairness are distinct criteria. An allocation can satisfy one without satisfying the other."

Template 3: Graph interpretation question (6–8 marks)

- 1 **Identify the diagram:** "The diagram shows [feasible frontier / indifference curves / Lorenz curve] with [free time / cumulative population] on the horizontal axis and [grain / income share] on the vertical axis."
- 2 **Identify the key curve/constraint:** "The [feasible frontier / IC1 / IC2] represents [maximum producible grain / Angela's reservation utility / Bruno's feasible set]."
- 3 **Locate the relevant point:** "At point [A/D/L], Angela has [X] hours of free time and receives [Y] bushels. Bruno receives [Z] bushels."
- 4 **Apply the MRS/MRT condition:** "At this point, the indifference curve is [tangent to / above / below] the feasible frontier. MRS [= / > / <] MRT, meaning [optimum is reached / a change in hours can increase surplus]."
- 5 **Draw the institutional conclusion:** "This outcome results from [Case X institution], which gives Bruno [type of power] and Angela [type of reservation option]. A change to [Case Y] would shift the allocation to [different point] because [reason]."

TUTORIAL APPLICATION

18. Tutorial 5 Problem Set — Worked Solutions

0% complete

Q1: Coercion vs voluntary exchange (from tutorial)

Given: total grain at 8h work = 8 bushels, at 11h work = 10 bushels.

Answer: Can Bruno be just as well off under voluntary exchange as under coercion? ▼

Q3: Jim (farmer) and Dwight (landowner)

Why does the biological survival constraint get steeper as Jim works more? ▼

What happens to (a) biological survival constraint and (b) feasible frontier if a more nutritious beet is available? ▼

Q4: $ATP = 16,000/(h+32) + 2h$ calculations

$$ATP = 16000/(h+32) + 2h$$

Q4.1: AP at $h=25$:

$$ATP = 16000/(25+32) + 2(25) = 16000/57 + 50 = 280.70 + 50 = \mathbf{330.70 \text{ units/hour}}$$

Q4.2: Total output $Q = h \times ATP = 16000h/(h+32) + 2h^2$

$$MP = dQ/dh = [16000(h+32) - 16000h]/(h+32)^2 + 4h$$

$$MP = \mathbf{51200/(h+32)^2 + 4h}$$

Q4.3: MP at $h=105$:

$$MP = 51200/(105+32)^2 + 4(105)$$

$$= 51200/(137)^2 + 420$$

$$= 51200/18769 + 420$$

$$= 27.28 + 420 = \mathbf{\approx 447.28 \text{ units/hour}}$$

Q5: Peter Parker (pomegranates) and Tony Stark

Q5.1: Which combinations are biologically infeasible?

Q5.2: If Tony offers 3 tons, how many hours does Peter work?

Q5.3: At point A, Tony offers 5 tons for 8 hours — should Peter accept?

DANGER
ZONE

19. Exam Traps & Negative-Marking Strategy

0% complete

#	The Trap Statement	Why it is WRONG	Correct statement
1	"Voluntary exchange is always fair"	Voluntariness does not imply equal power. Angela can refuse Bruno's offer in Case 2, but Bruno captures all the joint surplus as rent. The process may be formal but the power is unequal.	Voluntary exchange can still be unequal if one party has a much better reservation option and all the bargaining power.
2	"If Angela survives, the allocation is fair"	Survival only means Bruno respected the biological survival constraint. It says nothing about equality, consent, or procedural justice.	Angela surviving does not make the allocation fair — she receives far less than she would as an independent farmer.
3	"Pareto efficient allocations are fair"	Case 1 (forced labour, Angela gets 15, Bruno gets 31) is Pareto efficient. By any reasonable standard, it is profoundly unfair.	Pareto efficiency and fairness are completely different criteria. An allocation can be one without the other.
4	"Bruno maximises grain by making Angela work 24 hours"	At 24 hours work, MRT is very low (production function is nearly flat) but the survival constraint is high. The net gain to Bruno is lower than at 8 hours work.	Bruno maximises his rent where MRT (feasible frontier slope) = MRS (IC1 slope) — which is at 16 hours free time / 8 hours work.
5	"The legal minimum wage in Case 3 is Pareto efficient"	At the restricted work level (fewer than 8 hours), $MRT > MRS$. A Pareto improvement is available — both can gain from further bargaining.	The legal minimum in Case 3 is NOT Pareto efficient; further bargaining can reach a Pareto-efficient point.
6	"Market income Gini is lower than	Taxes and transfers reduce inequality. Disposable income is	Market income Gini EXCEEDS disposable income

#	The Trap Statement	Why it is WRONG	Correct statement
	disposable income Gini"	more equal (lower Gini). Market income is pre-redistribution (higher Gini).	Gini. The gap measures the extent of redistribution.
7	"The Lorenz curve shows how much is produced at each employment level"	The Lorenz curve is a distributional tool — it shows the cumulative income/wealth share of the population, not production levels.	The Lorenz curve measures the distribution of income or wealth, not production.
8	"Institutions determine how much a country can produce"	In the Angela/Bruno model, institutions (cases 1–3) keep the same feasible frontier — the same technology. Institutions change the allocation, not the production possibility.	In this model, institutions do not change the production technology (feasible frontier). They change who has power and how the fixed total is divided.

Negative-Marking Decision Framework

True/False (−½ for wrong)

Expected value of random guess: $1 \times \frac{1}{2} - \frac{1}{2} \times \frac{1}{2} = 0$. **Break-even at 50% confidence.**

Rule: Only answer if you have >55% confidence. If you know it is false-ish but can't pinpoint why → skip. If you can identify a specific error in the statement → answer.

MCQ (−¼ for wrong, 4 options)

Expected value of random guess among 4: $2 \times \frac{1}{4} - \frac{1}{4} \times \frac{3}{4} \times 4 \dots$ actually $EV = 2 \times (\frac{1}{4}) - \frac{1}{4} \times (\frac{3}{4}) = 0.5 - 0.1875 > 0$. It pays to guess even randomly.

Rule: Always attempt MCQ. If you can eliminate even one option, the EV of guessing rises further. Never leave MCQ blank.

ACTIVE RECALL

20. Active Recall — 40 Flash Cards

0% complete

Click each card to flip. Say the answer before flipping.

reifer (income) AND free time. Higher indifference curves = higher utility. She has quasi-linear preferences.

As much grain as possible. He does not care about Angela's free time. His ICs are horizontal lines.

The amount of grain Angela is willing to give up for one more hour of free time. Slope of the indifference curve.

The opportunity cost of one hour of free time — how much grain must be sacrificed. Slope of the feasible frontier = marginal product of labor.

When $MRS = MRT$ — the IC is tangent to the feasible frontier. In this model, always 16 hours free, 8 hours work, 48 bushels output.

Angela: 16h free, 8h work, 48 bushels (keeps all). Bruno: not present.

Angela: 16 bushels. Bruno: 31 bushels. Same hours (16h free). Angela has zero economic rent.

Angela: 23 bushels. Bruno: 23 bushels. Same hours. Angela still has zero economic rent. Bruno captures entire joint surplus.

YES. Any change from allocation D makes at least one party worse off. Pareto efficiency does NOT require fairness.

NO. At fewer than 8 hours work, $MRT < MRS$. A Pareto improvement is still available through further bargaining.

The benefit from a chosen action above the benefit of the next best alternative (reservation option). Angela has zero net benefit in her reservation IC.		Power from having a better fallback option — ability to walk away from a bad deal. Better outside option = more structural power.	
Sum of economic rents of all parties — total gain above all parties' reservation options. In Case 2: $48 - 23 = 23$ pushers above IC.		IC = Angela's reservation under coercion (escape/revolt). IC = Angela's reservation under voluntary exchange (alternative employment). IC is above IC — better outside option.	
Substantive: judged by the outcome (how equal is it?). Procedural: judged by the process (was it voluntarily legitimated?).		The bottom x% of the population earns y% of total income. The further this is below the 45° line, the more unequal.	
Perfect equality — every % of population earns the same % of income. The Lorenz curve coincides with the 45° line. NOT necessarily rich or poor.		$AP = Y/h$. At $h=8, Y=48$: $AP = 48/8 = 5.75$ pushers per hour.	
$MP = \Delta Y/\Delta h$. Between $h=6$ ($Y=41$) and $h=8$ ($Y=48$): $MP = (48-41)/(8-6) = 7/2 = 3.5$ pushers per hour.		Angela's MRS at any given level of free time is the same on every indifference curve. ICs are vertical shifts of each other. Result: optimal hours are always 8 regardless of which case.	
The region between the feasible frontier (top boundary) and IC (bottom boundary). These are all allocations Bruno can enforce — Angela won't revolt and production is feasible.		Angela with consumption = output = 23 maximises at 8h work (where $MRS = MRT$ on her consumption frontier). Bruno sets rent = 23 to capture this. Outcome = allocation L in both cases.	

critical numbers for exam calculations). 8 hours → 48 pushers. 12 hours → 54 pushers. (From production function table).		A change that makes at least one person better off without making anyone else worse off. If ANY person is made worse off, it is NOT a Pareto improvement.	
Because of diminishing marginal product. When Angela already has a lot of free time, each extra hour of free time costs her more gain (she was at a steeper part of the production function). $MRT = slope$ of feasible frontier, which rises as we move right.		Even illegal/violent organisations need institutions (rules of the game) to coordinate behaviour and share gains. Rules determine who has power and what each person gets.	
NO. In Case 2, Angela is better off ($18 \rightarrow 23$ pushers) but Bruno is worse off ($31 \rightarrow 23$ pushers). Neither case Pareto dominates the other. The Pareto criterion cannot rank them.		Bruno's rent falls. A higher IC means Bruno must offer Angela more gain to get her to accept. The gap between the feasible frontier and IC (= joint surplus Bruno can capture) shrinks.	
(1) Who has power (structural and bargaining). (2) Total pay-offs available (size of joint surplus). (3) How the total is divided (distribution of surplus between parties).		Rawls' thought experiment: evaluate the rules of a society as if you did not know which position you would occupy (Angela or Bruno, rich or poor). Enables impartial fairness judgements.	
Angela should take LESS free time (work more). Each hour of work produces more gain than Angela needs to compensate for the lost free time. Moving left increases her utility.		Angela should take MORE free time. The gain she is willing to sacrifice for one extra hour of free time exceeds what she would actually have to give up (MRT). Moving right increases her utility.	
2 macro (initial/unemployment) — that is institutional model (Angela/Bruno). NO Unit 5, 6, 7, 8, and 10. Unit 5 = micro		income BEFORE taxes and government transfers. Usually more unequal. Market income $Gini < income$ Disposable income Gini in countries with progressive tax systems.	

Quasi-linear preferences: MRS at any free-time level is the same across all ICs. So MRS = MRT is always satisfied at 16 hours free time \ 8h work, regardless of which IC or constraint is binding.

The process: voluntary exchange (no coercion), equal opportunity, legitimate property rights, non-discrimination, freedom of return. TOI just whether the outcome is equal.

A curve showing compensation as good as the reservation option. Angela stays on or above this curve; Bruno must ensure she reaches it (otherwise she returns/reverts). In Case 1 = IC1; in Case 2 = IC2.

In Case 3, Angela gains some economic rent (positive) because the democratic legislation raises her reservation option AND she can now bargain. In Case 2 she has zero rent.

Diminishing returns: each additional hour of work adds less output than the previous hour (because land is fixed). The production function flattens as hours increase. This makes the feasible frontier bow outward.

A vertical line at 16 hours free time (8 hours work) — because quasi-linear preferences mean all Pareto-efficient allocations have the same hours. Only the grain distribution between Angela and Bruno varies along this line.

21. One-Page Cram Sheet

🔑 Key Numbers

- ◆ Baseline: Angela 46bu, 16h free
- ◆ Case 1: Angela 15bu, Bruno 31bu
- ◆ Case 2: Angela 23bu, Bruno 23bu
- ◆ Joint surplus Case 2 = 23bu
- ◆ AP at h=8: 5.75 bu/hr
- ◆ MP (h=6→8): 2.5 bu/hr

⚡ Core Conditions

- ◆ Optimum: $MRS = MRT$
- ◆ $MRS = \text{slope of IC}$; $MRT = \text{slope of FF}$
- ◆ If $MRT > MRS \rightarrow$ work less
- ◆ If $MRS > MRT \rightarrow$ work more
- ◆ Bruno maximises: on IC, where $MRT=MRS$

📊 Inequality

- ◆ Lorenz curve: distribution, not production
- ◆ $Gini = \text{Area A} / (\text{A}+\text{B})$
- ◆ Gini 0=equality, 1=max inequality
- ◆ Market income Gini > disposable Gini

🏛️ Cases Summary

- ◆ Baseline: Angela owns land, $MRS=MRT$ freely
- ◆ Case 1: Bruno coerces, Angela on IC1, zero rent
- ◆ Case 2: Voluntary, Angela on IC2, zero rent
- ◆ Case 3: Democracy, Angela gets positive rent
- ◆ All cases: hours = 8, output = 46 (at optimum)

⚠️ Top Traps

- ◆ Voluntary ≠ fair or equal
- ◆ Efficient ≠ fair (Case 1 is BOTH)
- ◆ Survival ≠ fair

- ◆ Legal min $\neq$ Pareto efficient (Case 3)
- ◆ Lorenz $\neq$ production curve
- ◆ Market Gini $>$ Disposable Gini

👉 TF/MCQ Rules

- ◆ TF: only answer if $>55\%$ confident
- ◆ MCQ: always guess ($EV > 0$ for 4 options)
- ◆ Longer Q: use DICOM template (Define, IC, Condition, Outcome, Mechanism)

80% READINESS GATE

22. Unit 5 — 80% Readiness Checklist

0% complete

You are ready for Unit 5 in the A2 only when you can tick ALL of these:

Final check:

If you can tick all 12 items above and have completed the TF, MCQ, and flash card sections with $<20\%$ error rate, you are Unit 5 ready. Move on to the other A2 units. Come back to Unit 5 two days before the exam to review the cram sheet.